Reproducibility review of: Investigating the Generalizability of Segment Anything Model for Large-Scale Geospatial Segmentation

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Reviewed paper

TODO ADD FULL CITATION

Summary

- Workflow with environment packages in Docker container
- demanding system requirements (GPU, storage), largely undocumented
- CURRENT STATUS: No small enough subset working for reproduction

Reproducibility reviewer notes

The article comprises an analysis of the performance of different machine learning models. It provides a clear Data and Software Availability section that points to the following resources, which are the starting point for the reproduction:

> *Both datasets are open source (Werner et al., 2023; Li et al., 2021). The code and preprocessed datasets are available at the following links:*
>
> - Code repository: https://github.com/tum-bgd/2026-AGILE-GeoFM • BBD: https://figshare.com/s/83c2cead0d8fcc65dc6c • Surface Water Dataset: https://figshare.com/s/a359b71447c18d02b9db

I forked the code repository to https://github.com/reproducible-agile/2026-AGILE-GeoFM at commit `0deeb785ac77cf4a27f2ae3f4334b6e9822b9970` to preserve the reproduction state and cloned the fork locally. Then I followed the instructions in the README.

The commands from section **Download Checkpoints for GroundingDINO, RemoteCLIP and RemoteSAM**

The clone operation from **Download RemoteSAM GitHub Repository** completes without error, though I'm not sure I clone it to the right location, because the README does not specify where to clone it to, and RemoteSam is not in the `.gitignore` of the project.

In section **Data** the README properly references the original datasets, but for the preprocessed/cleaned data it, the links point to private records on Figshare without a DOI. The preprocessing is *not* checked. The preprocessing script for *Surface Water* is provided in the project, for *BBD* there seems to be no script but only a minimal description of manual (?) cleaning steps. The download, however, works, and I save the zip files in the `/data` directory and unzip them there with some manual adjustments because of different file structures, which hopefully results in the expected folder structure. The existing files `water1k_data_filtered.csv` and `bbd1k_data_filtered.csv` are overwritten by the extraction.

While the downloads are running, I start building the Docker image as documented in **Environment** with `docker build -t python-geofm .` (because of local issues with Docker I needed to add `--network=host`) and the build runs through.

When I try to run the provided command to start the container, I get an error related to the GPU:

```
docker run --gpus device=1 --name geofm-container -d -it --mount type=bind,source="$(pwd)",target=/GEO python-geofm
7390c4411bab427bd66562aadfbacee806e144a131a78c575495a2f8932ccedd
docker: Error response from daemon: failed to create task for container: failed to create shim task: OCI runtime create failed: runc create failed: unable to start container process: error during co
nvidia-container-cli: initialization error: load library failed: libnvidia-ml.so.1: cannot open shared object file: no such file or directory: unknown

Run 'docker run --help' for more information
```

Leaving out the `--gpus` option results in the container starting, but I am not sure I can run any of the scripts because of missing GPU support.

I continue:

```
docker run --name geofm-container -d -it --mount type=bind,source="$(pwd)",target=/GEO python-geofm
docker cp geofm-container:/GEOtmp/GroundingDINO ./GroundingDINO
Successfully copied 34.7MB to /home/daniel/git/reproducible-agile/reviews-2026/reports/018/2026-AGILE-GeoFM/GroundingDINO
```

In the commands for running the models, *it is unclear with parametrization is needed to conduct the same steps as in the original work based on the instructiosn in the README along.* The authors only refer to the details in the article and made small changes to the README.

I try the first command with an additional `base` option, hoping that will be quicker:

```
$ docker exec -it geofm-container bash
root@5fd6d10ef029:/GEO# 2026-AGILE-GeoFM [main]$ docker run --name geofm-container -d -it --mount type=bind,source=&quot;$(pwd)&quot;,target=/GEO python-geofm
bash: 2026-AGILE-GeoFM: command not found
root@5fd6d10ef029:/GEO# 5fd6d10ef02933c879d0f4c0448d9483c197714ca6354aa33fb8a41ee8a838a7
bash: 5fd6d10ef02933c879d0f4c0448d9483c197714ca6354aa33fb8a41ee8a838a7: command not found
root@5fd6d10ef029:/GEO#
root@5fd6d10ef029:/GEO# python3 src/sam_gt_prompt.py --dataset bbd1k --model_name large --prompt bb
total images: 4650
The device used is cpu
/opt/conda/lib/python3.10/site-packages/huggingface_hub/file_download.py:949: FutureWarning: `resume_download` is deprecated and will be removed in version 1.0.0. Downloads always resume when possib
  warnings.warn(
preprocessor_config.json: 100%|...| 466/466 [00:00&lt;00:00, 2.92MB/s]
config.json: 6.57kB [00:00, 15.8MB/s]
model.safetensors: 100%|...| 1.25G/1.25G [00:13&lt;00:00, 92.7MB/s]
  1%|...| 47/4650 [29:42&lt;45:11:35, 35.35s/it]
```

It seems to pick up the CPU - good - but if I read the progress bar right then the execution time is over 40 hours. That's not feasible.

Based on the information from the author that the `water1k` dataset is considerably smaller, I pick a specific command for that dataset, hoping it will be faster and allow me to achieve a partial reproduction. The following command should create parts of the data shown in Table 6 – see inset below the information by the authors on mapping parameters to tables.

```
python3 src/sam_dino_prompt.py --dataset water1k --model_name large --dino_model_name SwinT --text_prompt water --box_threshold 0.35 --text_threshold 0.25

FutureWarning: Importing from timm.models.layers is deprecated, please import via timm.layers
total images: 312
The device used is cpu
FutureWarning: `resume_download` is deprecated and will be removed in version 1.0.0. Downloads always resume when possible. If you want to force a new download, use `force_download=True`.
UserWarning: torch.meshgrid: in an upcoming release, it will be required to pass the indexing argument. (Triggered internally at /opt/conda/conda-bld/pytorch_1682343967769/work/aten/src/ATen/native/
final text_encoder_type: bert-base-uncased
tokenizer_config.json: 100%|...| 48.0/48.0 [00:00<00:00, 386kB/s]
config.json: 100%|...| 570/570 [00:00<00:00, 6.13MB/s]
vocab.txt: 232kB [00:00, 4.60MB/s]
tokenizer.json: 466kB [00:00, 4.36MB/s]
model.safetensors: 100%|...| 440M/440M [00:04<00:00, 90.5MB/s]
  0%|
Traceback (most recent call last):
  File "/GEO/src/sam_dino_prompt.py", line 140, in <module>
    main(args)
  File "/GEO/src/sam_dino_prompt.py", line 67, in main
    boxes, _, _ = predict(
  File "/GEOtmp/GroundingDINO/groundingdino/util/inference.py", line 64, in predict
    model = model.to(device)
  File "/opt/conda/lib/python3.10/site-packages/torch/nn/modules/module.py", line 1145, in to
    return self._apply(convert)
  File "/opt/conda/lib/python3.10/site-packages/torch/nn/modules/module.py", line 797, in _apply
    module._apply(fn)
  File "/opt/conda/lib/python3.10/site-packages/torch/nn/modules/module.py", line 797, in _apply
    module._apply(fn)
  File "/opt/conda/lib/python3.10/site-packages/torch/nn/modules/module.py", line 797, in _apply
    module._apply(fn)
  [Previous line repeated 3 more times]
  File "/opt/conda/lib/python3.10/site-packages/torch/nn/modules/module.py", line 820, in _apply
    param_applied = fn(param)
  File "/opt/conda/lib/python3.10/site-packages/torch/nn/modules/module.py", line 1143, in convert
    return t.to(device, dtype if t.is_floating_point() or t.is_complex() else None, non_blocking)
  File "/opt/conda/lib/python3.10/site-packages/torch/cuda/__init__.py", line 247, in _lazy_init
    torch._C._cuda_init()
RuntimeError: Found no NVIDIA driver on your system. Please check that you have an NVIDIA GPU and installed a driver from http://www.nvidia.com/Download/index.aspx
```

The Python script does not seem to have an option to disable GPU support, and it seems to be hardcoded in the GroundingDINO code, which is not ideal for reproducibility. In my next attempt, I try to recreate some of the results from Table 7.

```
python3 src/sam_automatic_label.py --dataset water1k --model_name large --points_per_side 32 --label 'a surface water' --clip_model_name ViT-bigG-14 --clip_threshold 0.7

TODO
```

CONTINUE WITH HELP OF AUTHOR

Information from the authors provided on request on mapping parameters to figure tables

To generate the results, these are the commands run (the paper actually mentions all the parameters chosen for all the experiments):

Table1:

```
python3 src/sam_gt_prompt.py --dataset bbd1k --model_name base --prompt bb
python3 src/sam_gt_prompt.py --dataset bbd1k --model_name base --prompt center_pt
python3 src/sam_gt_prompt.py --dataset bbd1k --model_name base --prompt multiple_pts --nr_pts 5
```

For the rest of the table, just change –model_name to large and then huge

Table 2:

```
python3 src/sam_gt_prompt.py --dataset bbd1k --model_name large --prompt
multiple_pts --nr_pts 1
```

and for the rest of the table, you just change –nr_pts 1 to 2, 4, 5, 10, and 20.

Table 3:

For the grounded SAM results:

```
python3 src/sam_dino_prompt.py --dataset bbd1k --model_name large --dino_model_name
SwinT --text_prompt building --box_threshold 0.35 --text_threshold 0.25
```

And for the rest of the grounded SAM results, we replace `--text_prompt` with 'buildings', 'single buildings', 'building and structure', and 'building . buildings .'

For the rest of the table and to get the RemoteSAM results from Table 3:

```
python3 src/remote_sam_text_prompt.py --dataset bbd1k --text_prompt building
```

And for the rest of the RemoteSAM, we replace `--text_prompt` with 'buildings', 'single buildings', 'building and structure', and building buildings.

Table 4:

```
python3 src/sam_automatic_label.py --dataset bbd1k --model_name large --points_per_side
32 --label buildings --clip_model_name ViT-bigG-14 --clip_threshold 0.7
```

For the rest of the results from the table, change the number of grid points by changing `--points_per_side` to 64 and 128.

To try RemoteCLIP model, change `--clip_model_name` to ViT-b-32 or ViT-L-14.

For testing Tip-Adapter, the pretrained models are provided under data/bbd1k_cache:

```
python3 src/sam_automatic_label.py --dataset bbd1k --model_name large --points_per_side
32 --label buildings --clip_model_name ViT-bigG-14 --clip_threshold 0.7
--few-shot --cache_dir data/bbd1k_cache/ --shots 8 --beta 5 --alpha 1
```

There is a script `generate_data_tip_adapter_f.py`, used to select images to train tip adapter and another script `train_tip_adapter_f.py` used to train the model.

Table 5:

Similar to Table 1 and 2, just change dataset to water1k. For `--prompt multiple_pts`, we investigate `--nr_pts` for 2, 5 and 10 only.

As for before and after filtering, the results from the script are for after filtering (because they're more interesting). If you also want to get, the results without any filtering, you'll have to change the script: remove lines 139, 140 and 141 and only leave 142 and unindented.

Table 6:

Exact same as Table 3, just change dataset to water1k and text_prompt to the ones from the table.

Table 7:

Exact same as Table 4, just change dataset to water1k and label to 'a surface water'.

Recommendations to the authors

- In the README on GitHub:
  - Include files sizes in the documentation about downloading models and data, to help users with limited bandwidth or storage to make informed decisions about what to download; use wget commands with progress information
  - Include processing times (including efforts that went into preprocessing) to manage expectations of users with limited time or computational resources, and to make preprocessing efforts more transparent
  - Provide scripted instructions for downloading and unzipping data, even from Figshare, to make sure the folder/file structure is exactly as needed
  - Add information about how the figures and statistics in the article are connected to the scripts generating them
- Provide a smaller sample dataset to evaluate your work without downloading 8+ GB of data
- Add proper metadata to the Figshare datasets and refer to them via the DOI, not use privately shared records without a DOI

```
command line code
```

Colophon

This document was created using R Markdown using the following environment.

```
sessionInfo()
```

```
## R version 4.5.3 (2026-03-11)
## Platform: x86_64-pc-linux-gnu
## Running under: Ubuntu 22.04.5 LTS
##
## Matrix products: default
## BLAS:   /usr/lib/x86_64-linux-gnu/blas/libblas.so.3.10.0
## LAPACK: /usr/lib/x86_64-linux-gnu/lapack/liblapack.so.3.10.0  LAPACK version 3.10.0
##
## locale:
##  [1] LC_CTYPE=en_US.UTF-8       LC_NUMERIC=C
##  [3] LC_TIME=de_DE.UTF-8        LC_COLLATE=en_US.UTF-8
##  [5] LC_MONETARY=de_DE.UTF-8    LC_MESSAGES=en_US.UTF-8
##  [7] LC_PAPER=de_DE.UTF-8       LC_NAME=C
##  [9] LC_ADDRESS=C               LC_TELEPHONE=C
## [11] LC_MEASUREMENT=de_DE.UTF-8 LC_IDENTIFICATION=C
##
## time zone: Europe/Berlin
## tzcode source: system (glibc)
##
## attached base packages:
## [1] stats     graphics  grDevices datasets  utils     methods   base
##
## loaded via a namespace (and not attached):
##  [1] compiler_4.5.3    fastmap_1.2.0     cli_3.6.6         tools_4.5.3
##  [5] htmltools_0.5.9   otel_0.2.0        rstudioapi_0.18.0 yaml_2.3.12
##  [9] bspm_0.5.7        rmarkdown_2.31    knitr_1.51        xfun_0.57
## [13] digest_0.6.39     rlang_1.2.0       evaluate_1.0.5
```